

Matplotlib Lab Program — Q35 & Q36

Question 35: Scatter Plot Comparing Two Subject Marks

Aim

To write a Python program to draw a scatter plot comparing the marks scored by students in two subjects (Mathematics and Science).

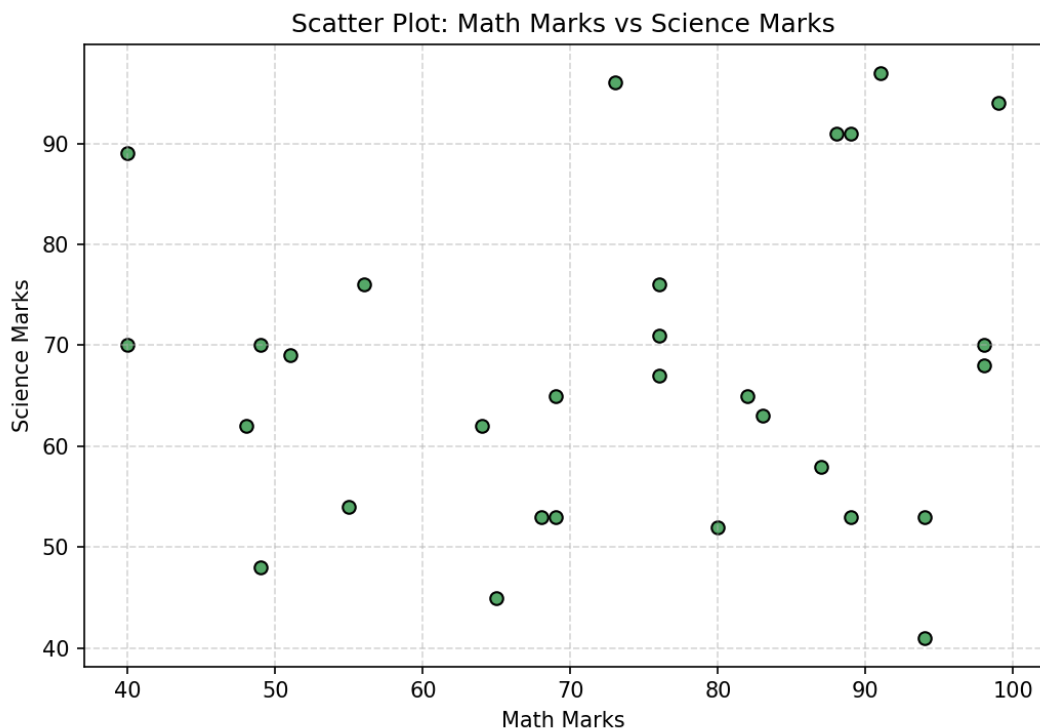
Source Code

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(10)
math_marks = np.random.randint(40, 100, 30)
science_marks = np.random.randint(40, 100, 30)

plt.figure(figsize=(7,5))
plt.scatter(math_marks, science_marks, color='#55A868', edgecolors='black')
plt.title('Scatter Plot: Math Marks vs Science Marks')
plt.xlabel('Math Marks')
plt.ylabel('Science Marks')
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig('imgs/q35_output.png', dpi=150)
print("Scatter plot comparing two subject marks created and saved successfully.")
```

Output



Result

The Python program was executed successfully and a scatter plot comparing Math marks against Science marks for 30 students was generated, showing the relationship between the two subjects.

Question 36: Scatter Plot Comparing Weights and Heights of Three Groups

Aim

To write a Python program to draw a scatter plot for three different groups, comparing their weights and heights.

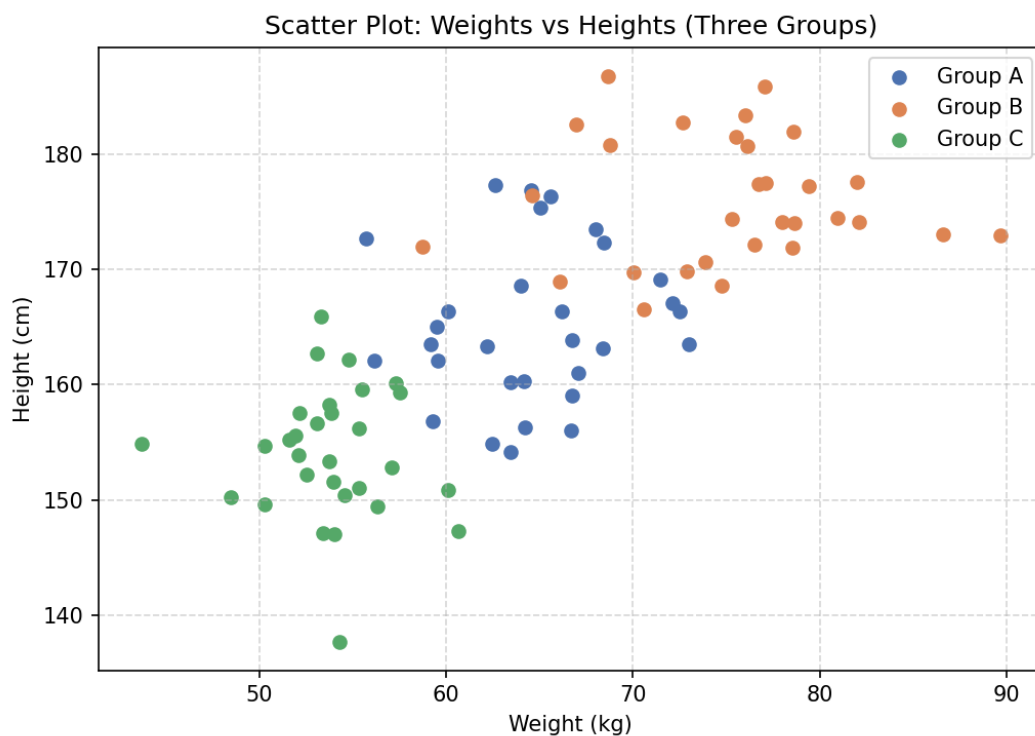
Source Code

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(15)
weight1, height1 = np.random.normal(65,5,30), np.random.normal(165,6,30)
weight2, height2 = np.random.normal(75,6,30), np.random.normal(175,5,30)
weight3, height3 = np.random.normal(55,4,30), np.random.normal(155,5,30)

plt.figure(figsize=(7,5))
plt.scatter(weight1, height1, color='#4C72B0', label='Group A')
plt.scatter(weight2, height2, color='#DD8452', label='Group B')
plt.scatter(weight3, height3, color='#55A868', label='Group C')
plt.title('Scatter Plot: Weights vs Heights (Three Groups)')
plt.xlabel('Weight (kg)')
plt.ylabel('Height (cm)')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig('imgs/q36_output.png', dpi=150)
print("Scatter plot comparing weights and heights for three groups created and saved successfully.")
```

Output



Result

The Python program was executed successfully and a scatter plot was created showing the weight vs. height distribution of three different groups (A, B and C), each represented with a distinct color and legend.